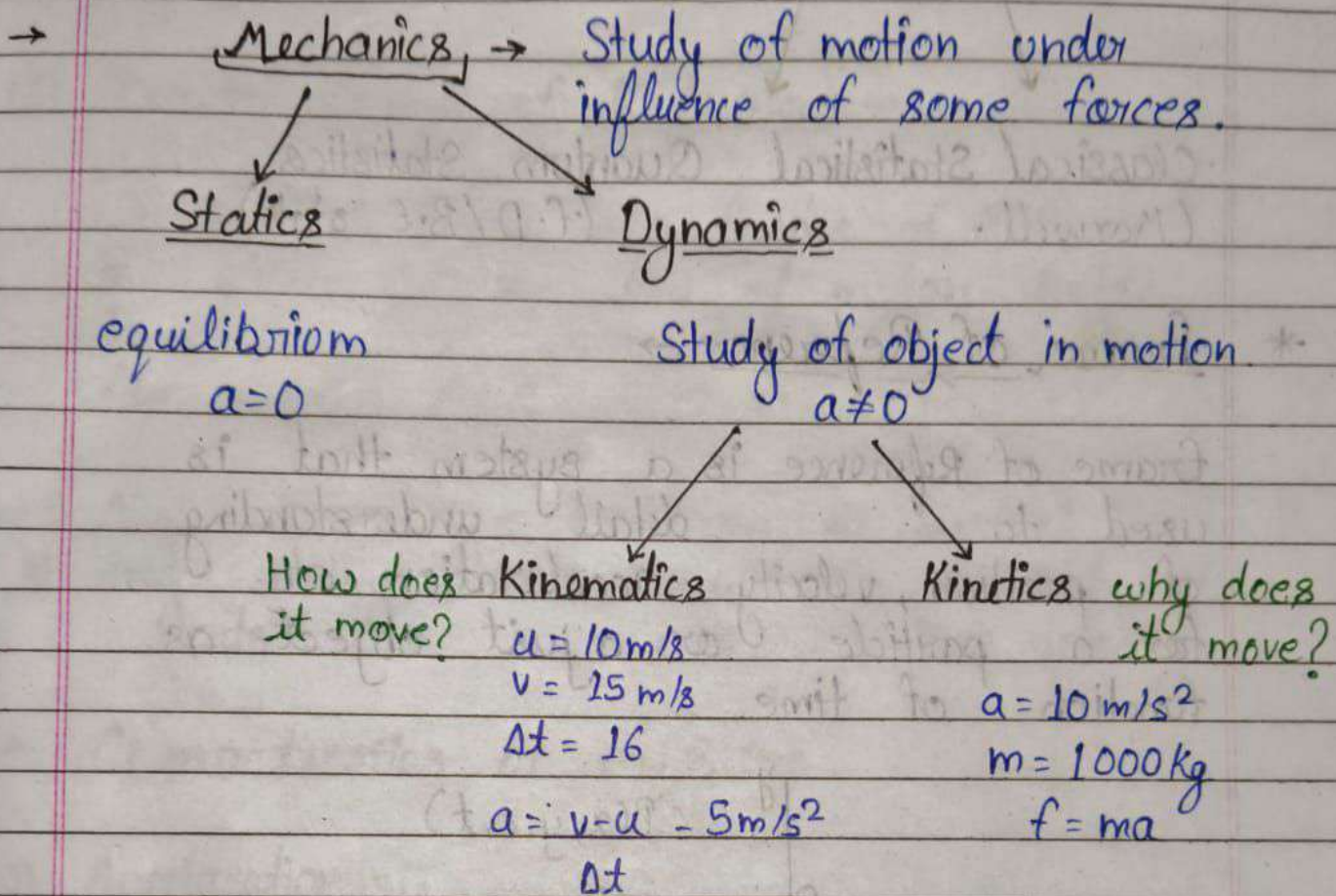


Relativistic Mechanics



① Classical Mechanics $\Rightarrow$ Study of macroscopic particles.

- * Newtonian Mechanics
- * Lagrangian Mechanics
- * Hamiltonian Mechanics

$V \ll c$
Non-Relativistic
C.M

$V \sim C$
Relativistic
C.M

② Quantum * Mechanics → Study of atomic or sub-atom particles.

V L L L C

$$V \sim C$$

Non-Relativistic Q.M

Relativistic Q.M

(schrodinger)

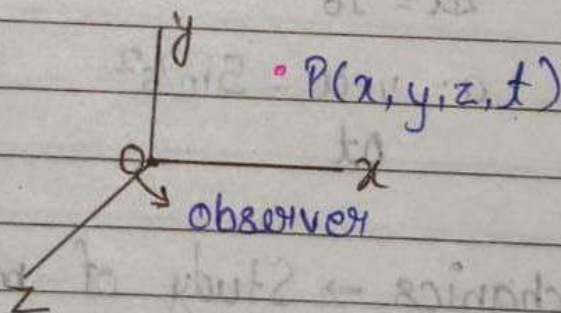
(Dirac, K.G.)

③ Statistical Mechanics $\rightarrow N \sim 10^{23}$

Classical Statistical (Maxwell -
Quantum Statistics (F.D/B.E stat)

* Frame of Reference $\rightarrow$

Frame of Reference is a system that is used to detail understanding of position, velocity, acceleration etc. for a particle or a just object as function of time.



A frame of reference is an arbitrary concept any observer can define its own frame of reference.

Ex $\rightarrow$ i) Any passenger moving in train its f.o.r is fixed inside the train.

ii) for train itself f.o.r can be one of Railway station.

iii) for motion of earth f.o.r can be on sun.

Frame of Reference can be in general of two types $\rightarrow$

① Inertial frame of Reference $\rightarrow$

An inertial f.o.r is a frame of Reference in which newton law of motion holds true.

for inertial f.o.r $a=0$

- Ex $\rightarrow$ i) Any stationary f.o.r can be inertial. *
- ii) Any moving f.o.r would be inertial if its with constant velocity.

* Characteristics of f.o.r $\rightarrow$

- i) Acceleration $= 0$
- ii) Uniformity of Laws of Mechanics.
- iii) Conservation of momentum.
- iv) Conservation of energy
- v) Maxwell Eqⁿ

Relative Uniform Motion $\rightarrow$

If any f.o.r is moving with constant velocity with another inertial f.o.r then it will also be inertial.

② Non-inertial f.o.r $\rightarrow$ In non-inertial f.o.r newton's law of motion does not hold true i.e. $a \neq 0$

Ex → i) Accelerated train or
Any rota frame can be non-inertial

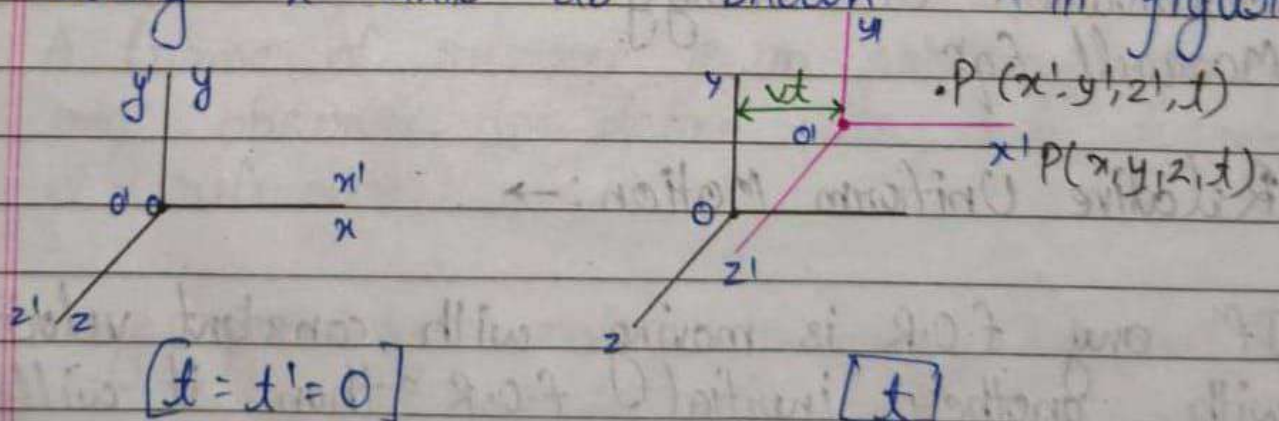
No frame of Reference is ideally **inertial** but for small scale experiment some frame of Reference.

Earth, moving train, moving space craft can be called inertial.

* Galilean Transformation →

Galilean transformation are used to establish to inertial frame of Reference.

Consider two f.o.r O and O' both are considering at any point in space at $t=0$ suppose the frame O' is allowed to move with velocity v along x axis as shown in figure →



from diagram we can write →

$$\left. \begin{aligned} x &= x' + vt \\ y &= y' \end{aligned} \right\}$$

$$\left. \begin{aligned} x' &= x - vt \\ y' &= y \end{aligned} \right\}$$

$$\left. \begin{matrix} z = z' \\ t = t' \end{matrix} \right\} \rightarrow \textcircled{1}$$

$$\left. \begin{matrix} z' = z \\ t' = t \end{matrix} \right\} \rightarrow \textcircled{2}$$

Eqⁿ ① and Eqⁿ ② are galilean transformation eqⁿ for position.

Now differential from Eqⁿ ② w.r.t t

$$\frac{dx'}{dt'} = \frac{dx}{dt} - \frac{d(vt)}{dt}$$

$$\frac{dy'}{dt'} = \frac{dy}{dt}, \quad \frac{dz'}{dt'} = \frac{dz}{dt}$$

$$\left. \begin{matrix} v'_x = v_x - v \\ v'_y = v_y \\ v'_z = v_z \end{matrix} \right\} \rightarrow \textcircled{3}$$

→ This is Gal^l eqⁿ velocity.

Again differentiating eqⁿ ③ w.r.t t

$$\frac{dv'_x}{dt'} = \frac{dv_x}{dt}$$

$$\frac{dv'_y}{dt'} = \frac{dv_y}{dt}$$

$$\frac{dv'_z}{dt'} = \frac{dv_z}{dt}$$

$$\left. \begin{aligned} a'_x &= a_x \\ a'_y &= a_y \\ a'_z &= a_z \end{aligned} \right\} \rightarrow (4)$$

Eqⁿ (4) is Gr eqⁿ for acceleration.

It is clear from eqⁿ (4) the acceleration under galilean transformation.

Galileon Invariant Quantities or Gr-invariance $\rightarrow$ which

The quantity, does not matter (change) under Gr.T are called galileon invariant quantity.

Mass, Time, acceleration, force etc.

Ques $\rightarrow$ Velocity of a particle in stationary frame is

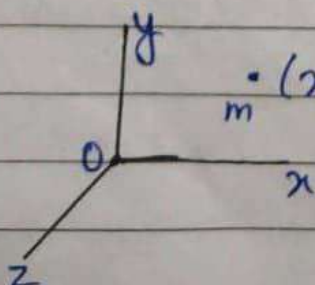
$$\vec{v} = 2\hat{i} + 3\hat{j} + 10\hat{k} \text{ m/s}$$

what would be its velocity if F.O.R moving with 5 m/s along z-axis?

$$\vec{v} = 2\hat{i} + 3\hat{j} + 5\hat{k}$$

Show the Galilean invariance of newton 2nd law.

Consider a stationary frame O-



$$m = (x, y, z, t) \quad \vec{F} = m\vec{a}$$

$$\vec{F} = m \left[\frac{d^2x}{dt^2} + \frac{d^2y}{dt^2} + \frac{d^2z}{dt^2} \right] \rightarrow$$

→ Suppose we have another frame moving with velocity (v) along x -axis. So, Galilean Transformation (G.T) are →

$$x = x' + vt$$

$$y = y'$$

$$z = z'$$

$$t = t'$$

→ from Eqⁿ ① and Eqⁿ ②, we can write →

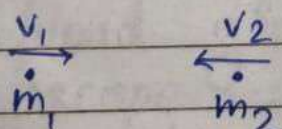
$$\vec{F} = m \left[\frac{d^2 x'}{dt'^2} + \frac{d^2 y'}{dt'^2} + \frac{d^2 z'}{dt'^2} \right]$$

$$[\vec{F} = m\vec{a}'] \rightarrow \text{③}$$

Since, the form of Newton's second law remains unchanged under Galilean transformation therefore, it is Galilean invariance.

Ques Show the invariance of conservation of momentum under G.T?

0 ——— x



$$m_1 v_1 + m_2 v_1 = m_1 u_1 + m_2 u_2$$

$$v_1' = v_1 - v \quad \left| \quad u_1' = u_1 - v \right.$$

$$v_2' = v_2 - v \quad \left| \quad u_2' = u_2 - v \right.$$

* Limitations of Galilean Transformation :→

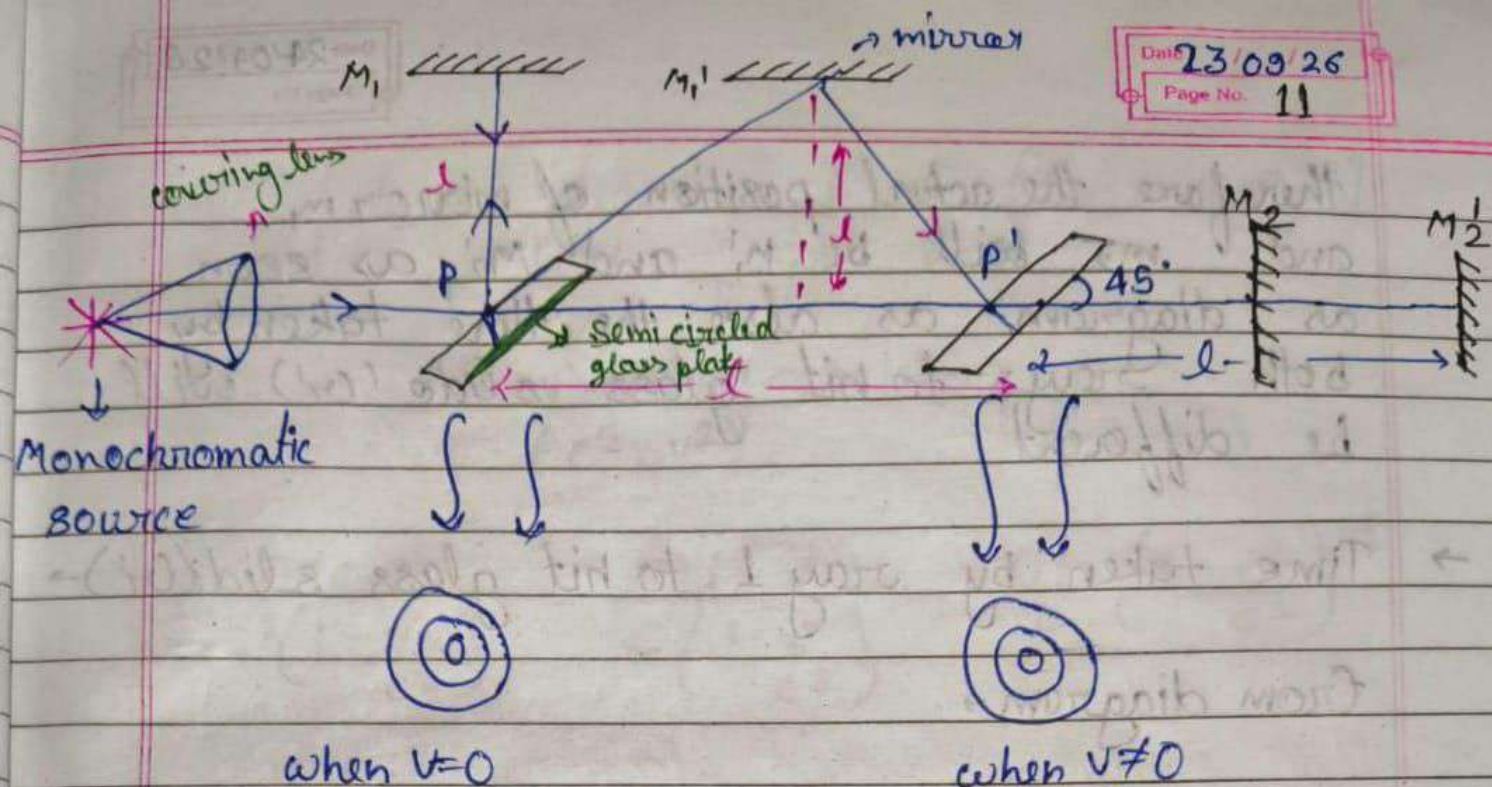
a) G.T assume absolute time and ordinary velocity addition. This idea looks for slow moving object. However, it has few limitations →

- b) It predict different speed of light in different frame of light. while it was well established that speed of light is constant (c) in all inertial frame.
- c) Gal. T would not provide appropriate transformation of maxwell eqn.
- (#) Michelson Morly $\rightarrow$
- (#) Aim of experiment $\rightarrow$

Michelson morly try to established an universal frame of reference which they call aether. this was supposed to be invisible, mass less, perfectly transparent, non-resistive having high-elasticity and low density. it was further suppose that earth moves through this intermedium and it is possible to determine its velocity by some experiment. the apparatus michelson morly use its called michelson-morly interferometer.

* Experimental Setup $\rightarrow$

the diagram of michelson-interferometer its shown $\rightarrow$ below $\rightarrow$



Light from monochromatic source (S) after being converge two lens falls on semi-circled silvered glass plate 'G' which is inclined at angle 45° to the beam from the direction of

It divide into two parts one reflected by semi silvered glass plate we named it ray ① travels towards mirror M_1 and other be transmitted through semi silvered glass be named it ray ② travels toward Mirror (M_2).

The two rays falls normally on mirror M_1 and M_2 and reflected back to the original path the reflected rays again beat at semi silvered glass plate and then inter in telescope where interference fringes are observed.

If apparatus is rest in aether. the two reflected rays will equal time to return to the semi silvered glass plate. however, the ^{whole} ~~total~~ apparatus is moving along earth therefore

therefore the actual position of mirror m_1 and m_2 will be m_1' and m_2' as shown in diagram as also the time taken by both rays to hit glass slide ' Gr ' will be different

→ Time taken by ray 1 to hit glass slide (Gr) →

from diagram →

$$(ct)^2 = (vt)^2 + l^2$$

$$c^2 t^2 = (v^2 t^2) + l^2$$

$$t^2 (c^2 - v^2) = l^2$$

$$t^2 = \frac{l^2}{(c^2 - v^2)}$$

$$t = \frac{l}{\sqrt{c^2 - v^2}} = \frac{l}{c \sqrt{1 - \frac{v^2}{c^2}}}$$

$$\left[t = \frac{l}{c} \left(\frac{1 - \frac{v^2}{c^2}}{1 + \frac{v^2}{c^2}} \right)^{-1/2} = \frac{l}{c} \left(\frac{1 + \frac{v^2}{c^2}}{1 - \frac{v^2}{c^2}} \right)^{1/2} \right]$$

Explaining using B.T. and ignoring higher order term.

Thus, time taken by Ray 1 to hit mirror M_1' and reflect back to Gr .

$$t_1 = 2t$$

$$\left[t_1 = \frac{2l}{c} \left(\frac{1 + \frac{v^2}{c^2}}{1 - \frac{v^2}{c^2}} \right)^{1/2} \right] \rightarrow \text{①}$$

Time taken by Ray 2 to hit mirror M_2' and reflect back to Gr →

$$t_2 = \frac{l}{c-v} + \frac{l}{c+v} \quad [\text{by O.T Eqn}]$$

$$t_2 = \frac{l(c+v) + l(c-v)}{(c^2 - v^2)} = \frac{2lc}{(c^2 - v^2)}$$

$$\Rightarrow \frac{2lc}{c^2 \left(1 - \frac{v^2}{c^2}\right)} = \frac{2l}{c \left(1 - \frac{v^2}{c^2}\right)} = \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right)^{-1}$$

$$\Rightarrow \boxed{t_2 = \frac{2l}{c} \left(1 + \frac{v^2}{c^2}\right)} \rightarrow \textcircled{2}$$

→ Therefore, time difference between Ray 1 and Ray 2 in reaching at glass slide (G_1').

$$\Delta t = \frac{2l}{c} \left(1 + \frac{v^2}{c^2}\right) - \frac{2l}{c} \left(1 + \frac{v^2}{2c^2}\right)$$

$$\Rightarrow \boxed{\Delta t = \frac{2l \cdot v^2}{c \cdot 2c^2} = \frac{lv^2}{c^3}} \rightarrow \textcircled{3}$$

Therefore, optical path difference between Ray ① and ② is equal to →

$$\Delta x = c \times \Delta t = c \times \frac{lv^2}{c^3}$$

$$\boxed{\Delta x = \frac{lv^2}{c^2}} \rightarrow \textcircled{4}$$

2. FitzGerald-Lorentz contraction hypothesis $\rightarrow$

According to this hypothesis length of moving object is contracted in direction of motion. if original length be l_0 then contracted length would be $\rightarrow l = l_0$

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

now $\rightarrow$

$$t_2 = \frac{2l}{c} \sqrt{1 - \frac{v^2}{c^2}} \cdot \left(1 + \frac{v^2}{c^2}\right)$$

$$= \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right)^{\frac{1}{2}} \left(1 + \frac{v^2}{c^2}\right)$$

$$= \frac{2l}{c} \left(1 - \frac{v^2}{c^2}\right) \left(1 + \frac{v^2}{c^2}\right)$$

$$= \frac{2l}{c} \left(1 + \frac{v^2}{c^2} - \frac{v^2}{c^2} - \frac{v^4}{c^4}\right)$$

$$= \frac{2l}{c} \left(1 - \frac{v^4}{c^4}\right) = t_1$$

$\rightarrow$ Hence, No fringe shift was observed.

$\rightarrow$ As this experiment was faulty and adhoc therefore not accepted.

$\rightarrow$ more over ^{when} the arm of interferometer was not equal this hypothesis failed to explain result.

3. Einstein Explanation :->

Einstein in 1905 proposed that motion of earth through ether is meaning less concept. the only motion relative to a frame of reference has its physical significance. the frame should be ~~word~~ earth, sun, any star but in every case you must be specify. this is reason why it is possible to ~~hit~~ detect motion of earth ~~theorem~~ ether.

* Importance of negative result of Michelson Morley Experiment ->

following conclusion can be drawn from the negative results of michelson morley.

1. The velocity of light is invalid, that is constant in all inertial frame of reference. hence, there is no meaning of $(c+v)$ and $(c-v)$.
2. Attempt to make ether as universal frame of Reference is meaning less.
3. A new theory having different space and needed new transformation relation is also needed in contrast with galilean transformation.

⊕ Postulates of special theory of Relativity :->

- i) Einstein proposed special theory of relativity in 1905.

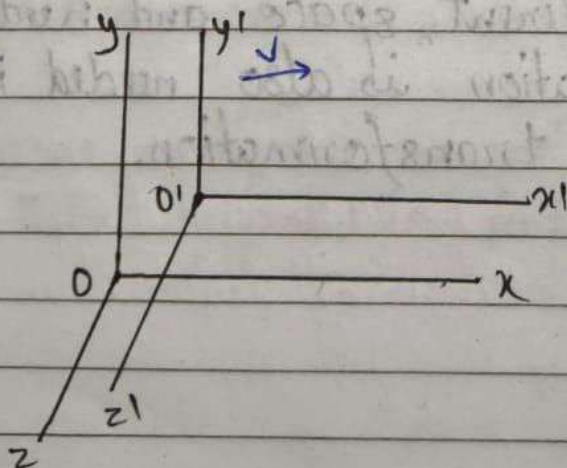
The postulates of special theory of relativity are ->

- i) All the laws of physics remain unchange or invariant in all inertial frame of Reference.
- ii) The speed of light is same (c) in all inertial F.O.R. It does not depend on the relative motion b/w frames.

⊕ Lorentz Transformation :->

-> Lorentz algebraically search a transformation relation which may replace transformation eqn to produce

According to michelson morley experiment -> These are called Lorentz transformation equation.



$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}; y' = y$$

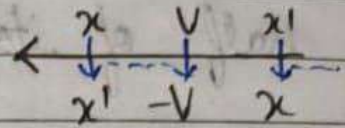
$$z' = z$$

$$t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

L.T.

The frame O' is moving with velocity (v) along x axis.

$$x = \frac{x' + vt'}{\sqrt{1 - \frac{v^2}{c^2}}}; y = y'; z = z'$$



$$t = \frac{t' - \frac{vx'}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \quad \text{I.L.T.}$$

Ques for Lorentz transformation check together $\rightarrow$ (whether)
 $x^2 + y^2 + z^2 - c^2 t^2 = x'^2 + y'^2 + z'^2 - c^2 t'^2$
 exist or not?

⑧ Length Contraction $\rightarrow$

The appeared decrease in the length of the body along the direction of motion. this called length contraction.



$$L < L_0$$

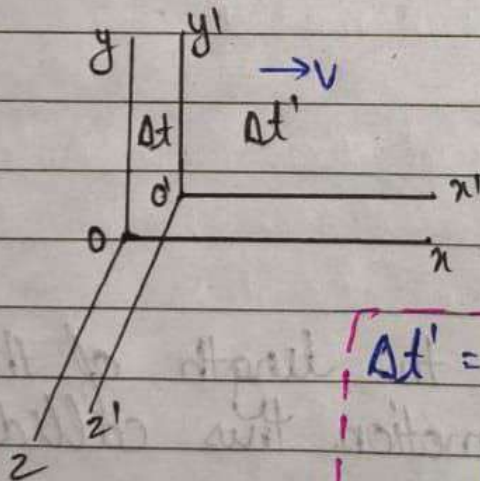
$$L = L_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Ques 2. At what speed the length of rod along its direction of motion, will appear to half of its original length?

Ans →

(*) Time dilation →

Moving clock appears to be running slower this is called time dilation.



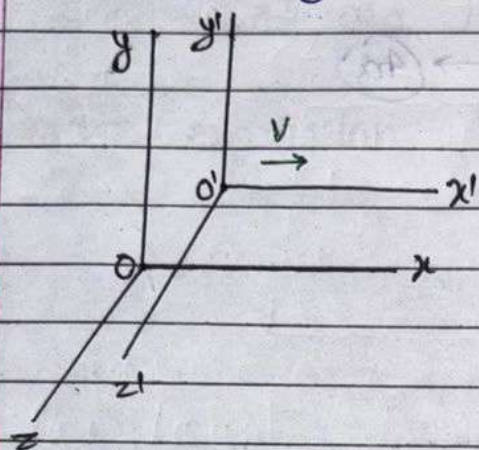
$$\Delta t < \Delta t'$$

moving runs slower
clock

$$\Delta t' = \frac{\Delta t}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Velocity Addition Theorem: →

Consider two frames O and O' where O' is moving with velocity (v) along x axis.



A/c to Lorentz transformation →

$$\left. \begin{aligned} x' &= \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}} \\ y' &= y \\ z' &= z \\ t' &= \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned} \right\} \rightarrow \textcircled{1}$$

$$y' = y, \quad z' = z, \quad t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$$

a point in O'

Velocity of frame along x axis can be given as →

$$U'_x = \frac{dx'}{dt'} \rightarrow \textcircled{2}$$

from eqⁿ ①, we can write →

$$\Rightarrow \begin{aligned} dx' &= \frac{dx - vdt}{\sqrt{1 - \frac{v^2}{c^2}}} \\ dt' &= \frac{dt - \frac{vdx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}} \end{aligned} \rightarrow \textcircled{3}$$

from eqⁿ ② and ③ →

$$\Rightarrow U'_x = \frac{dx - vdt}{\sqrt{1 - \frac{v^2}{c^2}}} \times \frac{\sqrt{1 - \frac{v^2}{c^2}}}{dt - \frac{vdx}{c^2}} = \frac{dx - vdt}{dt - \frac{vdx}{c^2}}$$

$$U'_x = \frac{dx/dt - v}{1 - \frac{v dx}{c^2 dt}} = \frac{U_x - v}{1 - \frac{v U_x}{c^2}}$$

 $\Rightarrow$

$$U'_x = \frac{U_x - v}{1 - \frac{v U_x}{c^2}} \rightarrow (4a)$$

Similarly $\rightarrow$

$$U'_y = \frac{dy}{dt'}$$

from eqⁿ ① $\rightarrow$

$$dy' = dy$$

and from eqⁿ ③ $\rightarrow$ $dt' = \frac{dt - \frac{v dx}{c^2}}{\sqrt{1 - \frac{v^2}{c^2}}}$

therefore,

$$U'_y = \frac{dy \times \sqrt{1 - \frac{v^2}{c^2}}}{dt - \frac{v dx}{c^2}}$$

$$U'_y = \frac{\frac{dy}{dt} \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v dx}{c^2 dt}} = \frac{U_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v U_x}{c^2}}$$

 $\Rightarrow$

$$U'_y = \frac{U_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v U_x}{c^2}} \rightarrow (4b)$$

$$\Rightarrow \boxed{U'_z = \frac{U_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_z v}{c^2}}} \rightarrow (4c)$$

→ Eqⁿ (4) (a, b, c) are rule for addition of velocities in a R.f.

Ques Write expression for additional velocity if frame O' is moving along y-axis, z-axis?

$$U'_x = \frac{U_x \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_y v}{c^2}} ; U'_y = \frac{U_y - v}{1 - \frac{U_y v}{c^2}}$$

$$U'_z = \frac{U_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_y v}{c^2}}$$

→ for z axis →

$$U'_x = \frac{U_x \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_z v}{c^2}} ; U'_y = \frac{U_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_z v}{c^2}}$$

$$U'_z = \frac{U_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_z v}{c^2}} = \frac{U_z - v}{1 - \frac{U_z v}{c^2}}$$

$$U'_x = \frac{U_x \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{U_y v}{c^2}} \longleftrightarrow U_x = \frac{U'_x \sqrt{1 - \frac{v^2}{c^2}}}{1 + \frac{U_y v}{c^2}}$$

(#) Variation of mass with velocity →

The mass of moving is very high speed relative to an observer is greater than rest mass. If m_0 is be the rest mass then mass

at speed $v \rightarrow$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

⑧ Einstein's mass-energy equivalence relation $\rightarrow$